CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

10.3 Potential Dividers

Syllabus Section: 10. D.C. Circuits **Paper Type:** Structured Questions (Paper 2) **Total:** 9 Questions (88 Marks)

Question 1 Easy [11 marks]

Part (a) [2 marks]

Sketch the circuits symbols for

(i) the potentiometer

[1]

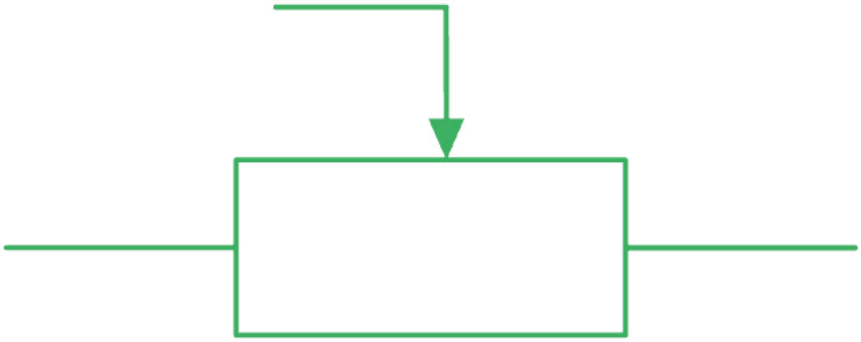
(ii) the galvanometer

[1]

Mark Scheme (2 marks):

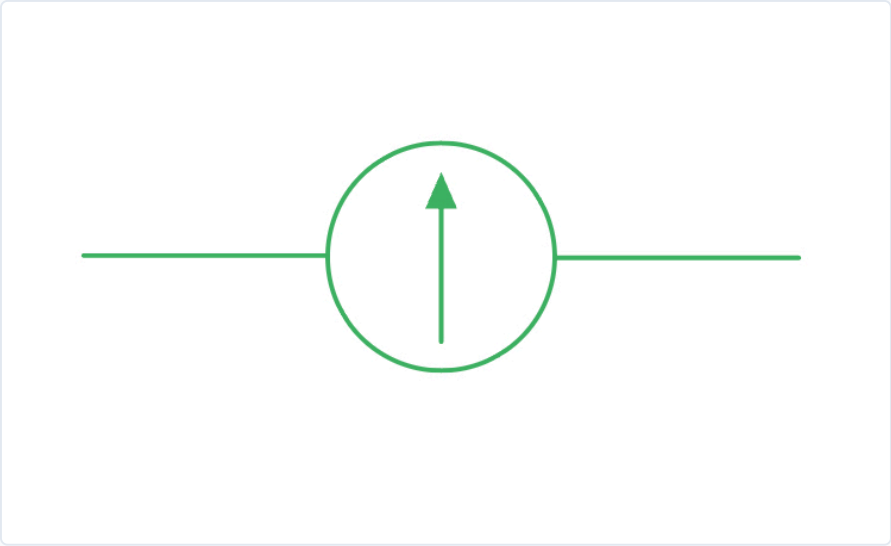
The circuit symbols for:

(i) the potentiometer is:



- Correct circuit symbol as shown above; *[1 mark]*

(ii) the galvanometer is:



- Correct circuit symbol as shown above; *[1 mark]*

[Total: 2 marks]

Part (b) **[3 marks]**

Fig. 1.1 shows a circuit with a potentiometer.

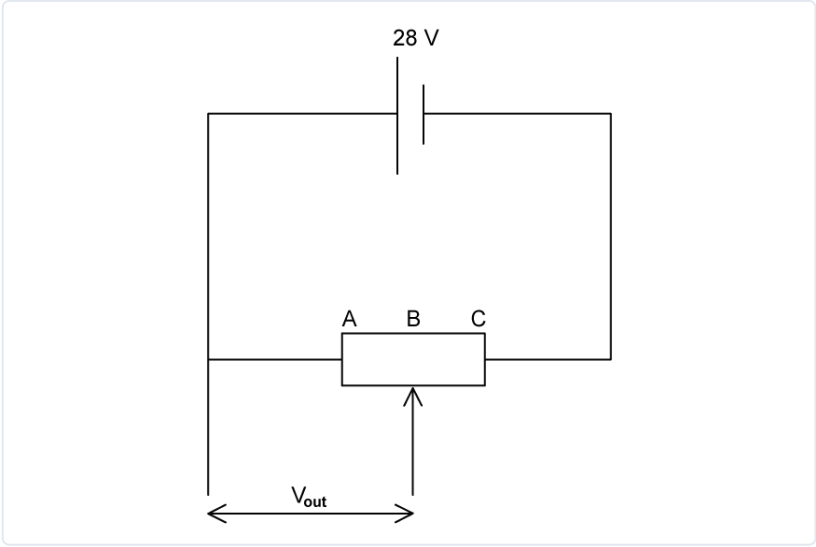


Fig. 1.1

Fixed ends AC are connected across the battery so that there is a full battery voltage across the whole resistor.

State the voltage output, V_{out} when the sliding connect is at

(i) A

[1]

(ii) B

[1]

(iii) C

[1]

Mark Scheme (3 marks):

The voltage output at:

(i) A is:

- 0 V *[1 mark]*

(ii) B is:

- 14 V *[1 mark]*

(iii) C is:

- 28 V *[1 mark]*

[Total: 3 marks]

Part (c) [2 marks]

The resistor is now replaced with a uniform resistance wire connected between ends AC, as shown in Fig. 1.2a. The 28 V cell is replaced with a cell of unknown e.m.f.

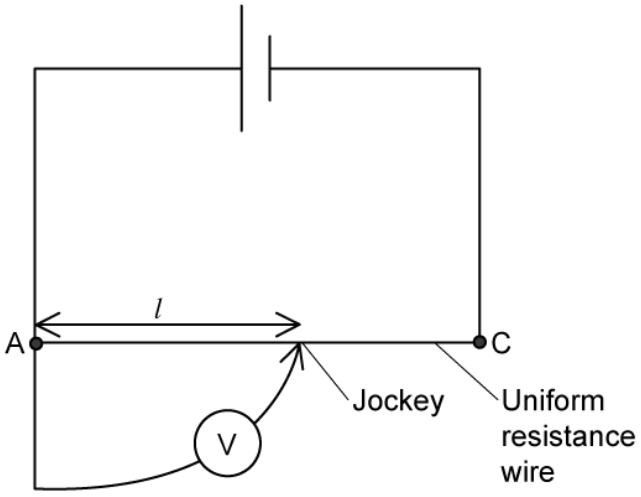


Fig. 1.2a

Sketch a graph of Fig 1.2b to show the voltmeter reading *V* against length *l* as the jockey is moved from point A to point C.

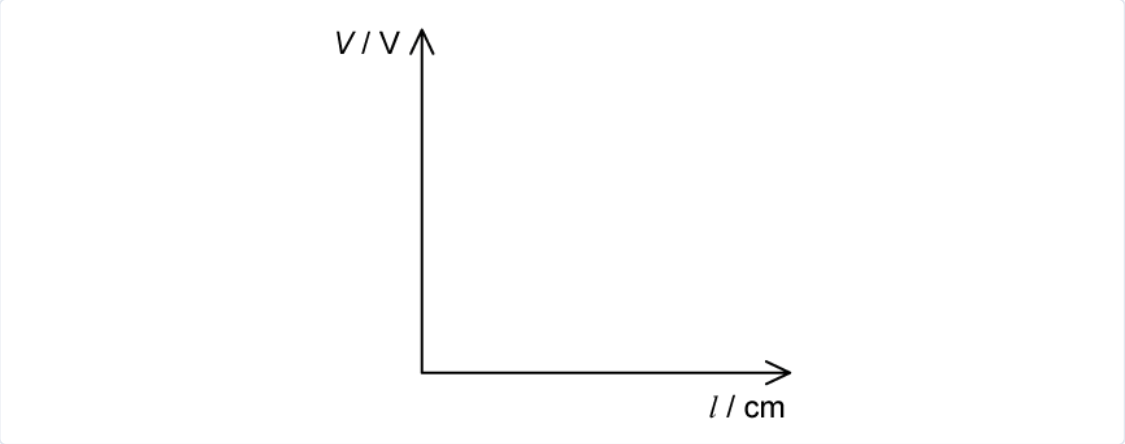
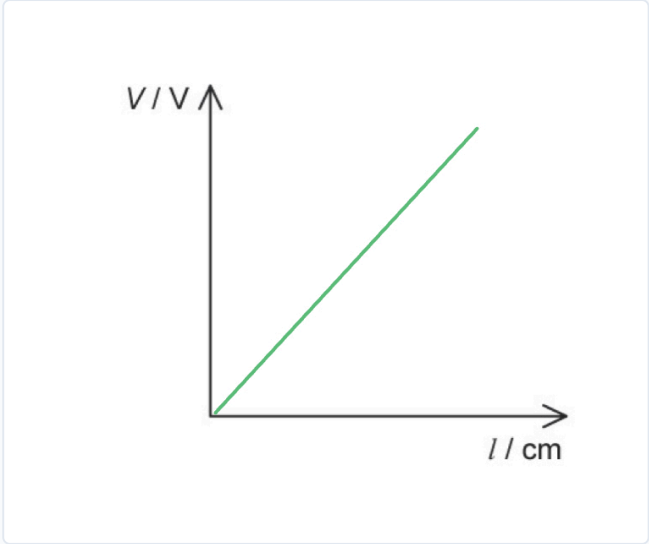


Fig 1.2b

Mark Scheme (2 marks):

A graph of the voltmeter reading *V* and length *l* of the jockey as it is moved from A to C is



- Straight line with a positive gradient; *[1 mark]*
- Through the origin; *[1 mark]*

[Total: 2 marks]

Part (d) **[4 marks]**

A potentiometer can be used to compare e.m.fs of cells or potential differences.

A test cell is added to the circuit in Fig. 1.2a. A student sets up the circuit shown in Fig. 1.3.

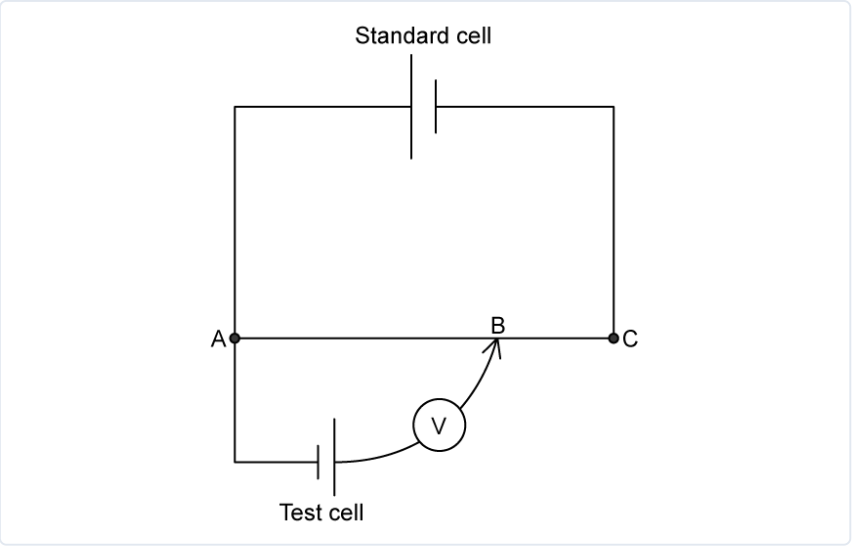


Fig. 1.3

(i) State why the student will not be able to find a point where the current is zero.

[1]

(ii) The student fixes the issue from **(a)(i)** and achieves a balance point when the length of the wire AB is 25.6 cm.

They repeat the experiment with a standard cell of e.m.f 1.545 V. The balance point using this cell is 37.2 cm.

Calculate the e.m.f of the test cell.

[3]

Mark Scheme (4 marks):

(i) The student will not be able to find a point where the current in zero because:

- The test cell is the wrong way around / the test cell needs to be reversed; *[1 mark]*

(ii) The e.m.f of the test cell is:

List the known quantities:

- Length AB at first balance point, l_1 = 25.6 cm
- E.m.f of standard cell, E_2 = 1.545 V
- Length AB at second balance point, l_2 = 37.2 cm

Determine the equation linking the length and e.m.fs:

- $\frac{l_1}{l_2} = \frac{E.m.f\ \ of\ \ test\ \ cell}{E_2}$ *[1 mark]*
- $E.m.f\ \ of\ \ test\ \ cell = E_2 \times \frac{l_1}{l_2}$

Substitute in the values:

- $E.m.f\ \ of\ \ test\ \ cell\ \ =\ \ 1.545\ \ \times \frac{25.6}{37.2}$ *[1 mark]*
- $E.m.f\ \ of\ \ test\ \ cell\ \ =\ \ 1.06\ \ \text{V}$ *[1 mark]*

[Total: 4 marks]

Question 2 Easy [8 marks]

Part (a) [2 marks]

Fig 1.1 shows a potential divider circuit containing a fixed resistor and thermistor.

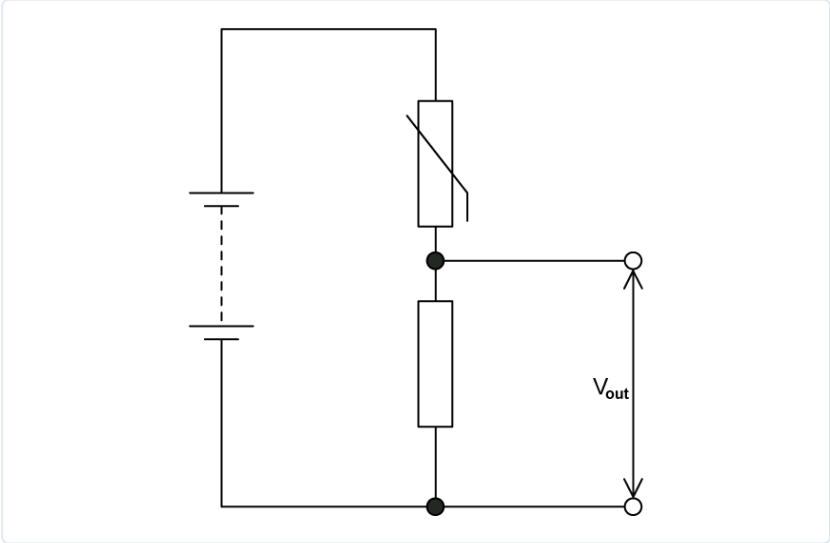


Fig 1.1

Explain what happens to V_{out} as the temperature of the thermistor increases.

Mark Scheme (2 marks):

As the temperature increases:

- The resistance of the thermistor decreases; *[1 mark]*
- So V_{out} increases; *[1 mark]*

[Total: 2 marks]

*Potential dividers are all about the **sharing** of the potential from the battery (hence the name). When the resistance of the thermistor decreases, this means the voltage through it also decreases (from V = IR where I is the same for both resistors).*

*However, the battery has a set voltage. Therefore, the voltage of the fixed resistor must **increase** so the combined voltage of the thermistor and fixed resistor remains the same. V_{out} is the reading of the voltage from the fixed resistor.*

Part (b) [1 mark]

The thermistor is now replaced with an LDR and the output voltage is now taken from the LDR, as shown in Fig. 1.2a.

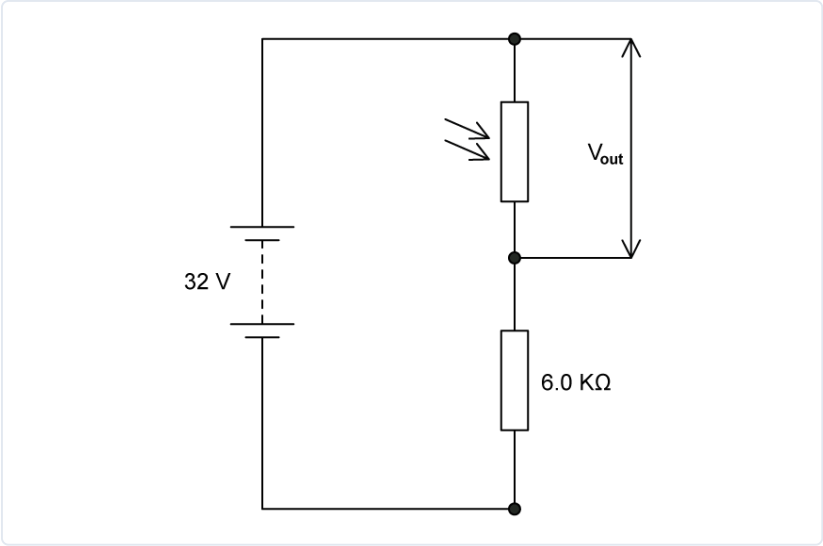


Fig. 1.2a

Using Fig. 1.2b, determine the resistance of the LDR when the light intensity is 120 lux.

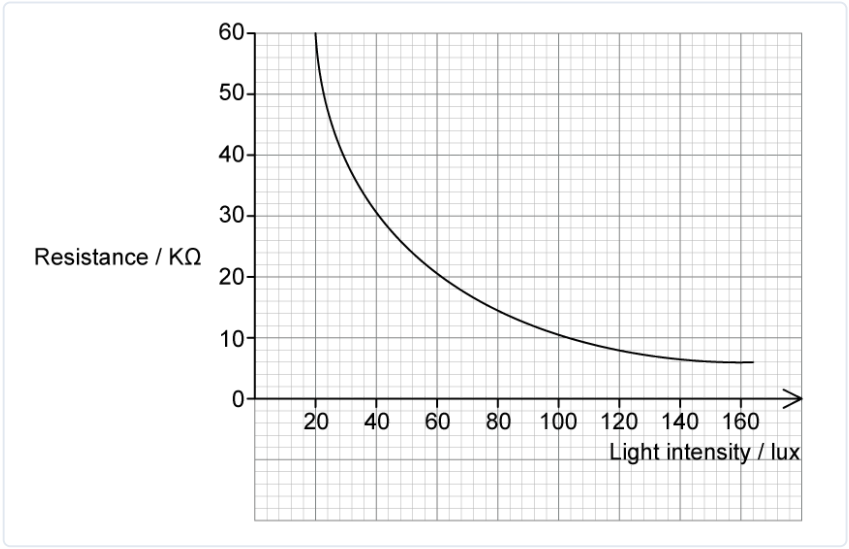
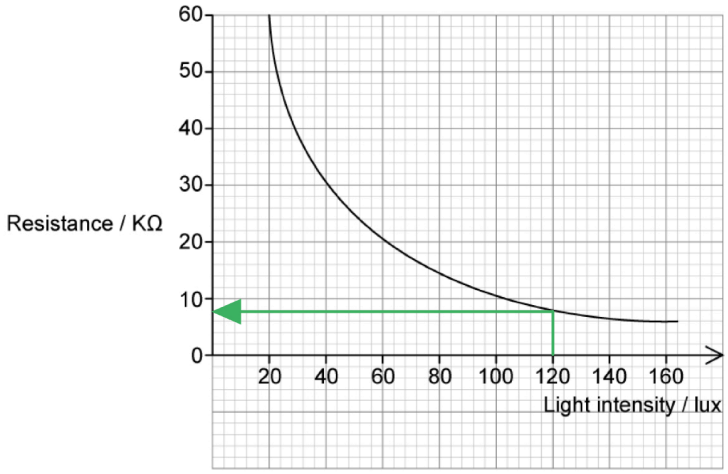


Fig. 1.2b

Mark Scheme (1 marks):

The resistance of the LDR when the light intensity is 120 lux is:



- 8 kΩ *[1 mark]*

[Total: 1 mark]

Part (c) **[3 marks]**

Using your answer to part **(b)**, calculate V_{out} when the light intensity is 120 lux.

Mark Scheme (3 marks):

V_{out} when the light intensity is 120 lux is:

List the known quantities:

- Resistance of the LDR, R_1 = 8.0 kΩ (from part b)
- Resistance of the fixed resistor, R_2 = 6.0 kΩ
- Voltage of the power supply, V_{in} = 32 V

Determine the equation for V_{out}:

- $V_{out} = \frac{R_1}{R_1 + R_2} \times V_{in}$ *[1 mark]*

Substitute in the values:

- $V_{out} = \frac{8.0}{8.0 + 6.0} \times 32$ *[1 mark]*
- $V_{out} = 18.2857 = 18$ V *[1 mark]*

[Total: 3 marks]

Part (d) **[2 marks]**

Using your answer to part **(c)**, determine the voltage across the 6.0 kΩ resistor at a light intensity of 120 lux.

Mark Scheme (2 marks):

The voltage across the 6.0 kΩ resistor is:

List the known quantities:

- Voltage across the LDR, V_{out} = 18 V
- Voltage of the power supply, V_{in} = 32 V

Calculate the voltage across the 6.0 kΩ resistor:

- $V_{6.0\text{k}\Omega} = V_{in} - V_{out} = 32 - 18$ *[1 mark]*
- $V_{6.0\text{k}\Omega} = 14$ V *[1 mark]*

[Total: 2 marks]

Question 3 Easy [8 marks]

Part (a) [4 marks]

A variable resistor is used to control the circuit shown in Fig. 1.1.

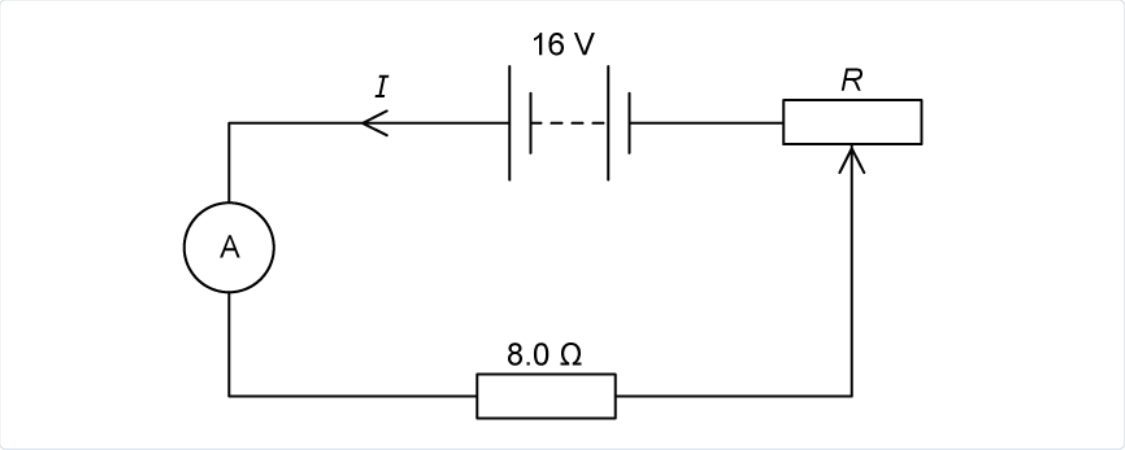


Fig. 1.1

The variable resistor is connected in series with a 16 V power supply of negligible internal resistance, an ammeter and an 8.0 Ω resistor.

The resistance *R* of the variable resistor can be varied between 0 to 12 Ω.

Calculate

(i) the minimum possible current

[2]

(ii) the maximum possible current

[2]

Mark Scheme (4 marks):

(i) Calculate the minimum possible current:

- $V = IR \Rightarrow I = \frac{V}{R}$
- $I \propto \frac{1}{R}$ so minimum current occurs with maximum resistance
- In series: R_{max} = 8 + 12 = 20 Ω *[1 mark]*

Calculate the minimum current:

- $I_{min} = \frac{16}{20}$
- $I_{min} = 0.8$ A *[1 mark]*

(ii) Calculate the maximum possible current:

- Maximum current occurs with minimum resistance, R_{min} = 8 Ω
- $I_{max} = \frac{16}{8}$ *[1 mark]*
- $I_{max} = 2.0 \text{ A}$ *[1 mark]*

[Total: 4 marks]

Part (b) [2 marks]

On Fig 1.2, sketch the variation with *R* of current *I* in the circuit.

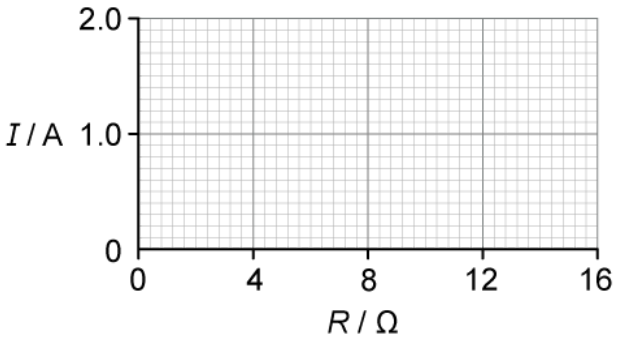


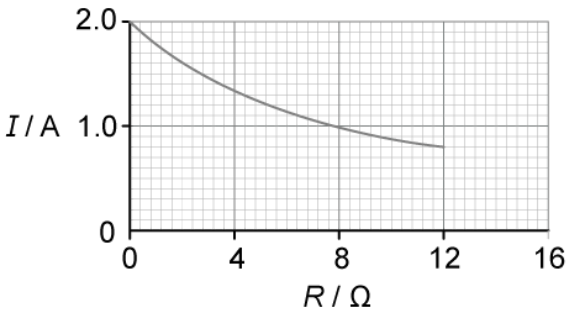
Fig. 1.2

Mark Scheme (2 marks):

The variation with *R* of current *I* in the circuit should look like:

List the known quantities:

- Maximum current = 2.0 A (from part a)
- Minimum current = 0.8 A (from part a)



- Starting point at (0, 2.0) and end point at (12, 0.8); *[1 mark]*
- Curve with decreasing gradient; *[1 mark]*

[Total: 2 marks]

Part (c) **[2 marks]**

Galvanometers are used in *null methods*.

Explain how a galvanometer is used in *null methods*.

Mark Scheme (2 marks):

Galvanometers are used in null methods by:

- The galvanometer has no deflection; *[1 mark]*
- When there is no current flow / no potential difference; *[1 mark]*

[Total: 2 marks]

Question 4 Medium [8 marks]

Part (a) [2 marks]

A battery is connected in series with resistors P and Q, as shown in Fig 1.1.

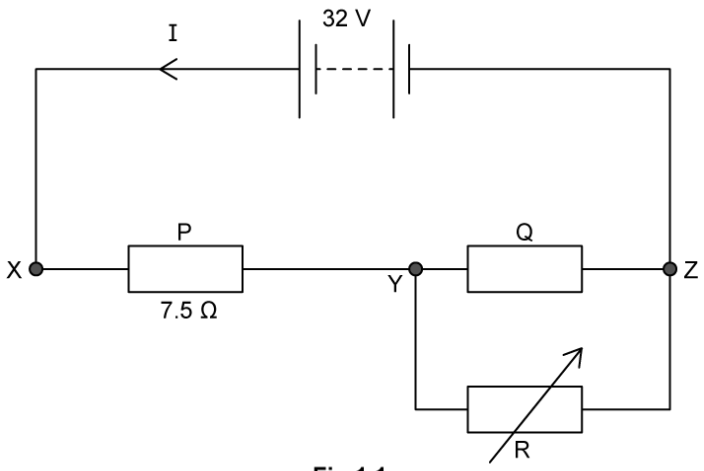


Fig.1.1

The resistance of P is 7.5 Ω. The resistance of Q is constant. A variable resistor of resistance *R* is connected in parallel with Q.

The current *I* from the battery is changed by varying *R* from 5 Ω to 20 Ω. The variation with *R* of *I* is shown in Fig 1.2.

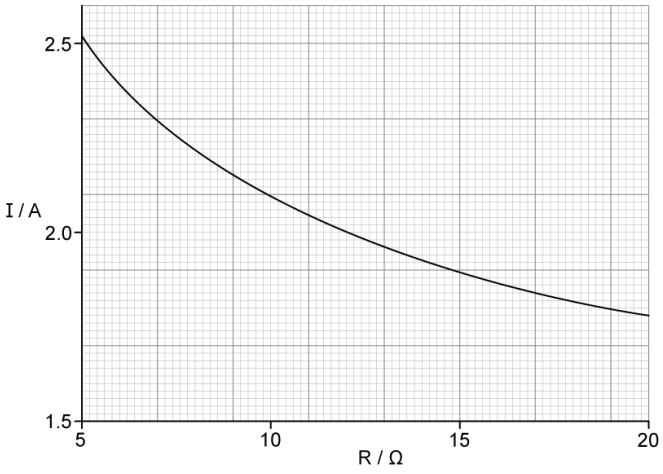


Fig.1.2

Use Fig 1.2 to state and explain the variation of the p.d. across resistor P as *R* is **decreased**. Numerical values are not required.

Mark Scheme (2 marks):

Use Fig 1.2 to state and explain the variation of the p.d. across resistor P as *R* is **decreased**

EITHER

- (As *R* decreases,) the current *I* increases; *[1 mark]*
- So the p.d. (across P) increases; *[1 mark]*

OR

- The parallel resistance of (of Q and *R*) decreases; *[1 mark]*
- The p.d. across the parallel resistors decreases, so p.d. across P increases; *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

For R = 7.0 Ω, calculate the p.d. between points Y and Z.

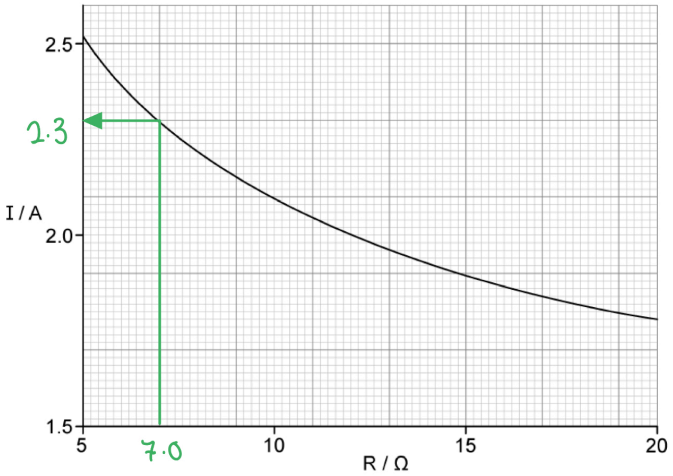
Mark Scheme (2 marks):

For R = 7.0 Ω, calculate the p.d. between points Y and Z

List the known quantities:

- Resistance of variable resistor, R = 7.0 Ω
- Resistance of resistor P, R_P = 7.5 Ω
- E.m.f of the power supply, V = 32 V

Use the graph to determine the current *I* when R = 7.0 Ω:



- I = 2.3 A *[1 mark]*

Method 1: Calculating p.d from current *I*

Calculate the p.d. across YZ:

- P.d across YZ = p.d across XZ – p.d. across XY

- P.d across YZ = 32 – (2.3 × 7.5) = 14.75 = 15 V *[1 mark]*

Method 2: Calculate p.d. from resistance

Determine the total resistance in the circuit, R_T:

- $R_T = \frac{V}{I} = \frac{32}{2.3} = 13.91 \ \Omega$

Calculate the p.d. across YZ:

- Resistance in Q-R parallel combination, R_{QR} = 13.91 – 7.5 = 6.41 Ω
- P.d across YZ = $I \times R_{QR}$ = 2.3 × 6.41 = 14.74 = 15 V *[1 mark]*

[Total: 2 marks]

Part (c) **[3 marks]**

Show that the resistance of Q is 95 Ω.

Mark Scheme (3 marks):

Show that the resistance of Q is 95 Ω

List the known quantities:

- Resistance of variable resistor, R = 7.0 Ω
- P.d. across YZ = 15 V (from part b)
- Current, I = 2.3 A (from part b)

Calculate the total resistance of Q and R parallel combination, R_T :

- $R_T = \frac{V}{I} = \frac{15}{2.3} = 6.52 \ \Omega$ *[1 mark]*

Calculate the resistance of resistor Q, R_Q :

- $\frac{1}{R_T} = \frac{1}{R_Q} + \frac{1}{R}$ *[1 mark]*
- $R_Q = \frac{1}{\frac{1}{R_T} - \frac{1}{R}} = \frac{1}{\frac{1}{6.52} - \frac{1}{7.0}} = 95 \ \Omega$ *[1 mark]*

[Total: 3 marks]

This question can also be done by calculating the current through Q, and then using $R = \frac{V}{I}$ to calculate the resistance through resistor Q.

Part (d) **[1 mark]**

State and explain qualitatively how the power provided by the battery changes as the resistance R is decreased.

Mark Scheme (1 marks):

State and explain qualitatively how the power provided by the battery changes as the resistance *R* is decreased

- The power increases

  AND

- Because the e.m.f is constant / power = *V* × current and the current (*I*) increases **OR** power = $\frac{V^2}{resistance}$ and the resistance decreases; *[1 mark]*

[Total: 1 mark]

Question 5 Medium [8 marks]

Part (a) [2 marks]

Fig 1.1 shows a potential divider circuit with a battery of electromotive force (e.m.f.) 24.0 V connected to two resistors R_1 and R_2.

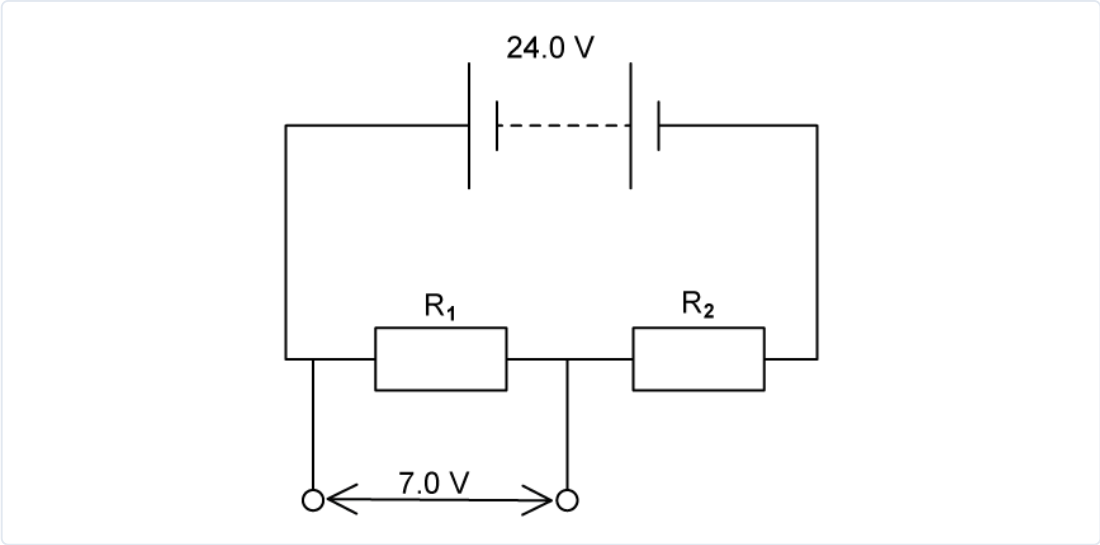


Fig 1.1

When the resistance of R_1 is 6.5 Ω, the voltage across it is 7.0 V.

Show that the resistance of R_2 is 16 Ω.

Mark Scheme (2 marks):

Show that the resistance of R_2 is 16 Ω

List the known quantities:

- Resistance, R_1 = 6.5 Ω
- Voltage across R_1, V_{out} = 7.0 V
- E.m.f of the power supply, V_{in} = 24.0 V

State the correct potential divider equation:

- $V_{out} = V_{in} \times \frac{R_1}{R_1 + R_2}$ *[1 mark]*

Rearrange the potential divider equation for R_2:

- $R_2 = \frac{R_1 \times V_{in}}{V_{out}} - R_1$

Substitute the values into the R_2 equation:

- $R_2 = \frac{6.5 \times 24.0}{7.0} - 6.5 = 15.786 = 16\ \Omega$ *[1 mark]*

[Total: 2 marks]

You must be confident with rearranging the potential divider equation. If you are struggling with this one, have a look at the sketch below:

$$V_{out} = V_{in}\left(\frac{R_1}{R_1+R_2}\right)$$

$$\frac{V_{out}}{V_{in}} = \left(\frac{R_1}{R_1+R_2}\right) \quad \downarrow \div V_{in}$$

$$\frac{V_{out}}{V_{in}}(R_1+R_2) = R_1 \quad \downarrow \times (R_1+R_2)$$

$$R_1+R_2 = \frac{R_1}{\frac{V_{out}}{V_{in}}} = \frac{R_1 V_{in}}{V_{out}} \quad \downarrow \div \frac{V_{out}}{V_{in}}$$

$$R_2 = \left(\frac{R_1 V_{in}}{V_{out}}\right) - R_1 \quad \downarrow -R_1$$

Part (b) [1 mark]

The resistors are replaced with identical filament lamps A and B and the battery is connected in parallel to a linear potentiometer XY, as shown in Fig 1.2.

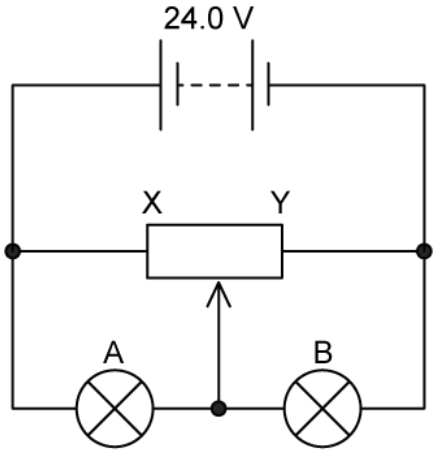


Fig 1.2

The lamps each have the same *I–V* characteristic shown in Fig. 1.3.

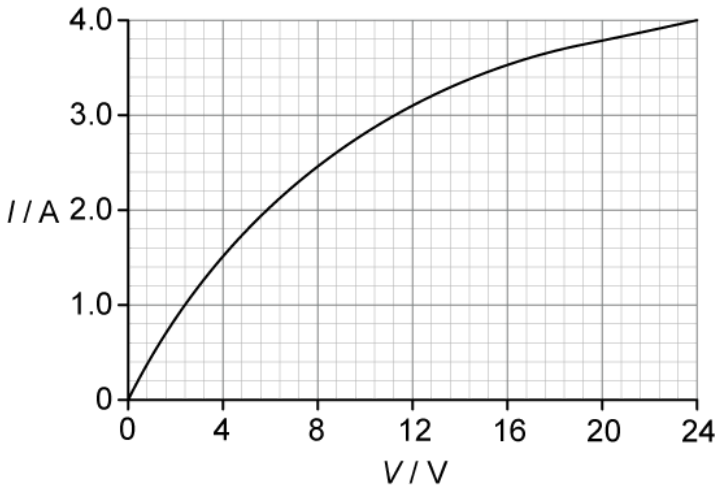


Fig 1.3

When the slider of the potentiometer is at its midpoint, as shown in Fig. 1.2, the current *I* in the battery is 4.12 A.

Determine the current in lamp B.

Mark Scheme (1 marks):

The current in lamp B is:

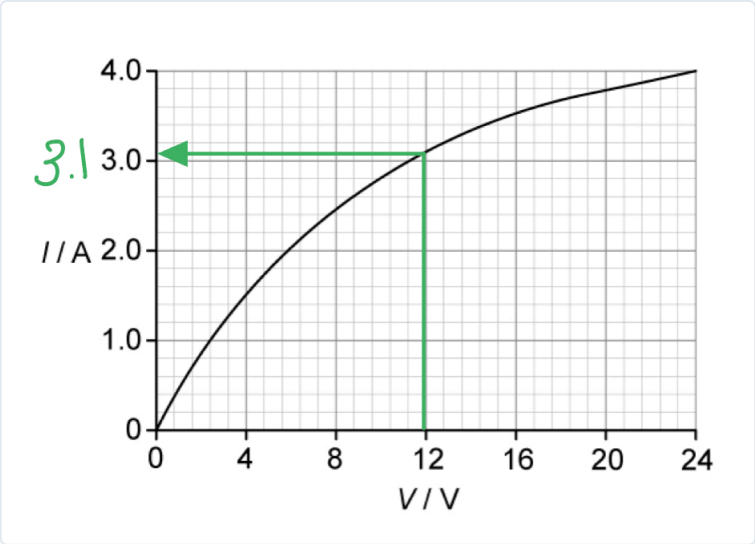
List the known quantities:

- Current in the battery, *I* = 4.12 A
- E.m.f of the battery = 24.0 V

Determine the potential difference through lamp B:

- Since the slider is exactly in the middle, both lamps have a potential difference of $\frac{24.0}{2} = 12 \text{ V}$

Determine the current through lamp B at 12 V from the graph:



- 3.10 A *[1 mark]*

[Total: 1 mark]

Part (c) **[3 marks]**

Calculate the total resistance in the circuit.

Mark Scheme (3 marks):

Calculate the total resistance in the circuit

List the known quantities:

- Current through lamp B, I_B = 3.10 A
- Current in the battery, I = 4.12 A
- E.m.f of the battery, V = 24.0 V

Calculate the resistance of the potentiometer between its ends X and Y, R_{XY}:

- Current through XY, $I_{XY} = I - I_B = 4.12 - 3.10 = 1.02$ A
- $R_{XY} = \frac{V}{I_{XY}} = \frac{24.0}{1.02} = 23.529\ \Omega$ *[1 mark]*

Calculate the resistance of each lamp, R_L:

- $R_L = \frac{V}{I} = \frac{12}{3.10} = 3.87\ \Omega$ *[1 mark]*

Calculate the combined resistance, R_T:

- $R_{series} = R_L\ lamp\ A + R_L\ lamp\ B = 3.87 + 3.87 = 7.74\ \Omega$
- $\frac{1}{R_T} = \frac{1}{R_{series}} + \frac{1}{R_{XY}} = \frac{1}{7.74} + \frac{1}{23.529}$
- $\frac{1}{R_T} = 0.1712$

- $R_T = 5.824 = 5.82\ \Omega$ *[1 mark]*

[Total: 3 marks]

*Read carefully which current the question is referring to. Since the potentiometer is connected in **parallel** with the lamps, the current from the **battery** will **split** between them (Kirchhoff's first law). Lamp A and B will have the same current through them.*

Part (d) **[2 marks]**

The slider of the potentiometer in (c) is moved to end X. State and explain the effect on the current of lamps A and B.

Mark Scheme (2 marks):

State and explain the effect on the current of lamps A and B when the slider of the potentiometer is moved to end X

For lamp A:

- The p.d. across lamp A decreases to zero **AND** so the current is 0 *[1 mark]*

For lamp B:

- p.d. across lamp B increases to 24.0 V **AND** so the current is 4.0 A *[1 mark]*

[Total: 2 marks]

*It is important you understand how the sliding contact works for a potentiometer. When it is at point X, **all** the p.d. from the power supply is available only to lamp B, so there is 0 for lamp A.*

*Where you **can** give values for a 'state and explain' question, make sure you do! In this case, just saying 'the p.d. across lamp B increases and so a current flows' will **not** be enough detail to obtain the marks, as you can state specific values.*

The 4.0 A is read from the I-V graph when lamp B has a voltage of 24 V.

Question 6 Medium [11 marks]

Part (a) [2 marks]

A battery of negligible internal resistance is connected to a light-dependent resistor, a switch X and two fixed resistors, as shown in Fig. 1.1.

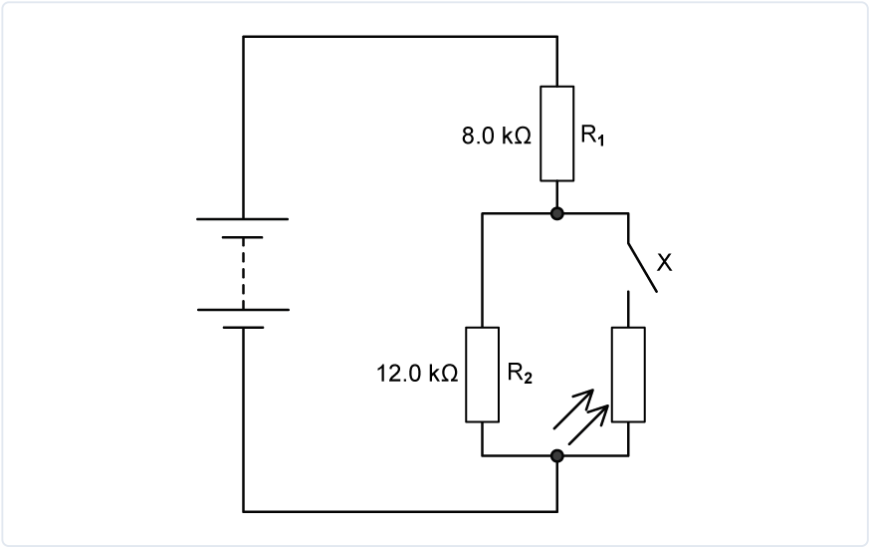


Fig 1.1

Resistor R_1 has resistance 8.0 kΩ and resistor R_2 has resistance 12.0 kΩ. When switch X is open, the potential difference across R_2 is 18 V.

Show that the e.m.f of the battery is 30 V.

Mark Scheme (2 marks):

Show that the e.m.f of the battery is 30 V

List the known quantities:

- Potential difference across R_2, V_{out} = 18 V
- Resistance of R_1, R_1 = 8.0 kΩ
- Resistance of R_2, R_2 = 12.0 kΩ

State the potential divider equation:

- $V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$ *[1 mark]*

Rearrange for the e.m.f of the battery, V_{in} and substitute in the values:

- $V_{in} = \frac{V_{out}}{\frac{R_2}{R_1 + R_2}} = \frac{18.0}{\frac{12.0}{8.0 + 12.0}}$ = 30 V *[1 mark]*

[Total: 2 marks]

You do not need to convert from kΩ into Ω, since the fraction $\frac{R_2}{R_1 + R_2}$ will have no units (they cancel out) as long as they're all in kΩ.

Part (b) [3 marks]

Switch X is now closed. The current in the battery is 1.94 mA.

Calculate the resistance of the LDR.

Mark Scheme (3 marks):

Calculate the resistance of the LDR.

List the known quantities:

- Resistance of R_1, R_1 = 8.0 kΩ
- Resistance of R_2, R_2 = 12.0 kΩ
- Current in the battery, I = 1.94 mA = 1.94 × 10^{-3} A
- E.m.f of the battery, V = 30 V (from part a)

Calculate the total resistance in the circuit, R_T:

- $R_T = \frac{V}{I} = \frac{30}{1.94 \times 10^{-3}} = 15.46$ kΩ *[1 mark]*

Calculate the resistance of the LDR, R_{LDR}:

- $15.46 = R_{parallel} + 8.0 \Rightarrow R_{parallel} = 15.46 - 8.0 = 7.46$ kΩ
- $\frac{1}{R_{parallel}} = \frac{1}{R_2} + \frac{1}{R_{LDR}} \Rightarrow \frac{1}{7.46} = \frac{1}{12.0} + \frac{1}{R_{LDR}}$
- $R_{LDR} = \frac{1}{\frac{1}{7.46} - \frac{1}{12.0}}$ *[1 mark]*
- $R_{LDR} = 19.7 = 20$ kΩ *[1 mark]*

[Total: 3 marks]

You should be confident in combining resistors in parallel and in series. It is better to set out the values algebraically before substituting in values.

Part (c) [3 marks]

The switch X in the circuit in (a) remains closed. The light intensity on the LDR increases.

By reference to the current in the battery, state and explain the effect, if any, of the increase in light intensity on the power produced by the battery.

Mark Scheme (3 marks):

State and explain the effect, if any, of the increase in light intensity on the power produced by the battery

- The LDR resistance decreases; *[1 mark]*
- (So the total resistance decreases) so the current increases (in the battery); *[1 mark]*
- Therefore, (since $P = IV$ and V is constant) so the power increases; *[1 mark]*

[Total: 3 marks]

Part (d) [3 marks]

The light intensity on the LDR decreases.

State and explain what happens to the potential difference across R_1.

Mark Scheme (3 marks):

If the light intensity on the LDR decreases:

- The resistance of the LDR increases; *[1 mark]*
- The combined resistance of the 12.0 kΩ / R_2 and the LDR increases; *[1 mark]*
- Therefore, the potential difference across R_1 decreases / R_1 receives a smaller share of the potential difference; *[1 mark]*

[Total: 3 marks]

The key to understanding potential divider questions is - if the resistance increases then so does the share of the potential difference (of the power supply) to that component.

Question 7 Medium [9 marks]

Part (a) [1 mark]

State Kirchhoff's second law.

Mark Scheme (1 marks):

Kirchhoff's second law states:

- The sum of the electromotive force(s) is equal to the sum of the potential difference(s) around a (closed) loop *[1 mark]*

[Total: 1 mark]

Part (b) [2 marks]

A battery with negligible internal resistance is connected in series with two resistors of resistances R_1 and R_2, as shown in Fig. 7.1.

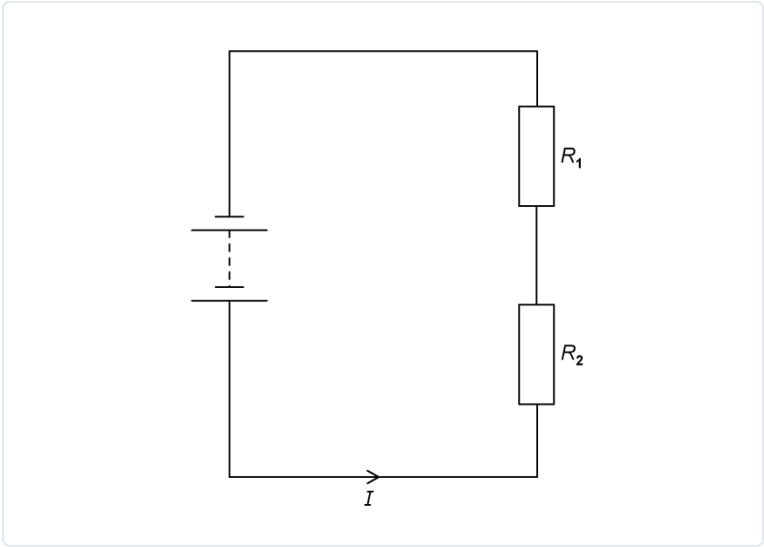


Fig. 7.1

The potential differences (p.d.s) across R_1 and R_2 are V_1 and V_2, respectively.

The p.d. across the combination of resistors is V.

The current in the circuit is I.

Use Kirchhoff's laws to show that the total resistance R_T of the combination is given by

$$R_T = R_1 + R_2$$

Mark Scheme (2 marks):

To derive an expression for the total resistance R_T of the combination:

- Use Kirchhoff's second law to write an expression for the total p.d. across the combination
- The total p.d. across the resistors = the sum of the p.d.s across each resistor
  - $V = V_1 + V_2$ *[1 mark]*
- Use Ohm's law to determine an expression for the resistances
  - Ohm's law: $V = IR$

Since, the current is the same in a series combination of resistors

- $IR_T = IR_1 + IR_2$ so $R_T = R_1 + R_2$

OR

- $\frac{V}{T} = \frac{V_1}{T} + \frac{V_2}{T}$ so $R_T = R_1 + R_2$ *[1 mark]*

[Total: 2 marks]

Part (c) **[3 marks]**

The resistor R_2 in Fig. 7.1 is replaced by a light-dependent resistor (LDR). The battery has an electromotive force (e.m.f.) of 12.0 V. The resistor R_1 has a fixed resistance of 2.40 kΩ.

A voltmeter is connected across the fixed resistor, as shown in Fig. 7.2. The reading on the voltmeter is 9.00 V.

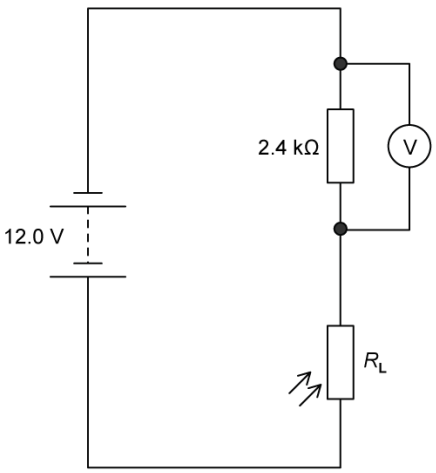


Fig. 7.2

(i) Calculate the current in the circuit.

[1]

(ii) Calculate the resistance R_L of the LDR.

[2]

Mark Scheme (3 marks):

List the known quantities:

- e.m.f. of the battery, E = 12.0 V
- p.d. across the fixed resistor, V_1 = 9.00 V
- Resistance of the fixed resistor, R_1 = 2.40 kΩ = 2400 Ω
- Resistance of the LDR, $R_2 = R_L$

(i) To calculate the current in the circuit:

$$I = \frac{V_1}{R_1}$$

$$I = \frac{9.00}{2400}$$

$$I = 3.75 \times 10^{-3} \text{ A } \textit{[1 mark]}$$

(ii) To calculate the resistance R_L of the LDR:

- Recall the potential divider equation:

$$V_1 = \frac{R_1}{R_1 + R_2} E \textit{ [1 mark]}$$

- Substitute the known quantities into the equation to calculate R_L

$$\frac{9.0}{12.0} = \frac{2400}{2400 + R_L}$$

$$R_L = \frac{2400 \times 12.0}{9.0} - 2400$$

$$R_L = 800 \text{ Ω } \textit{[1 mark]}$$

[Total: 3 marks]

Part (d) **[3 marks]**

The intensity of the light illuminating the LDR decreases.

By reference to the current in the circuit, state and explain the change, if any, to the voltmeter reading.

Mark Scheme (3 marks):

As the light intensity decreases, the voltmeter reading:

- Decreases *[1 mark]*

Explanation:

- The resistance of the LDR / R_L increases *[1 mark]*
- The current (in the circuit) decreases (so the p.d. across R_1 decreases) *[1 mark]*

[Total: 3 marks]

Question 8 Hard [16 marks]

Part (a) [3 marks]

A circuit is set up as shown in Fig. 1.1 below:

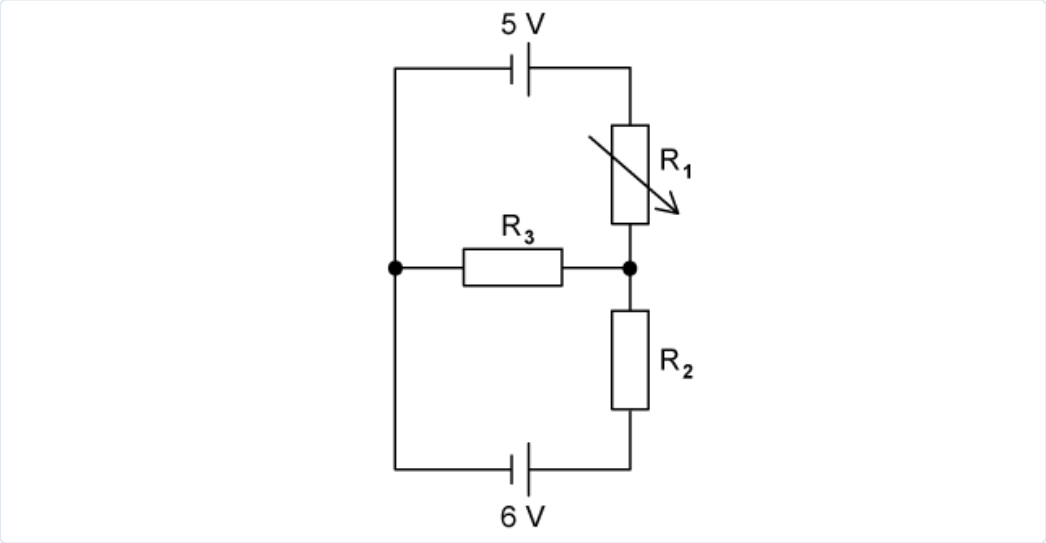


Fig. 1.1

The cells in Fig. 1.1 have negligible internal resistance. The cell at the top has an e.m.f. of 5 V and the cell at the bottom has an e.m.f. of 6 V.

R_1 is a variable resistor with a resistance that varies between 0 and 10 Ω, and R_2 and R_3 are fixed resistors with a resistance of 10 Ω and 30 Ω respectively.

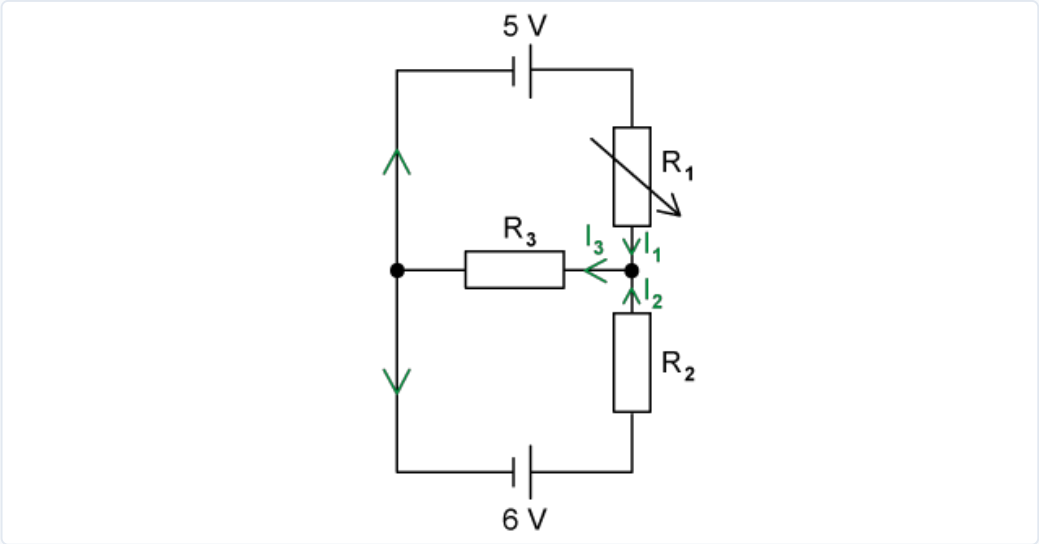
Initially, R_1 is set such that its resistance is 0 Ω.

By labelling the direction of current on Fig. 1.1, write down the relationship between three currents I_1 , I_2 and I_3 at the junction between R_1 and R_2 .

Mark Scheme (3 marks):

Label the direction of currents on Fig. 1.1:

- Labels on Fig. 1.1 which would score full marks are shown below:



- Arrows indicating correct directions for currents in both loops shown; *[1 mark]*
- Three distinct currents labelled at the junction between R_1 and R_2 as shown (any subscript is acceptable as long as distinct); *[1 mark]*

Write down the relationship between the currents at the junction between R_1 and R_2 :

- Using Kirchhoff's first law, the sum of currents into a junction must equal the sum of currents out of the junction
- Therefore, $I_1 + I_2 = I_3$ (or any equation consistent with correct labelling on Fig. 1.1); *[1 mark]*

[Total: 3 marks]

When you see 'junction', this should be a hint to apply Kirchhoff's first law. No calculation is needed at this stage, the question only asks for an expression and a labelled diagram.

Part (b) **[3 marks]**

Determine the current through the resistor R_2 .

Mark Scheme (3 marks):

List the known quantities:

- R_1 = 0 Ω
- R_2 = 10 Ω
- R_3 = 30 Ω

Apply Kirchhoff's second law to the top loop:

- Kirchhoff's second law states that, in a closed loop, the total e.m.f. is equal to the sum of potential differences
  - In the top loop, e.m.f. is 5 V so $\varepsilon = 5 = V_1 + V_3$

- V_1 must be 0 V, as resistor 1 has no resistance so there is no change in potential across it
    - Therefore $V_3 = 5$ V *[1 mark]*

Apply Kirchhoff's second law to the bottom loop:

- In the bottom loop, e.m.f. is 6 V so $\varepsilon = 6 = V_2 + V_3$
    - V_3 is 5 V so $6 = V_2 + 5$
    - $V_2 = 1$ V *[1 mark]*

Calculate the current through resistor R_2:

- Using Ohm's law across R_2 : $V_2 = I_2 R_2$
- Therefore, 1 = I_2 × 10
- $I_2 = \frac{1}{10}$ = 0.10 A *[1 mark]*

[Total: 3 marks]

Equally, you could have applied the second law to the bottom loop then the top loop, or a loop around the whole circuit. All should produce the same potential differences across each resistor.

Part (c) **[4 marks]**

State and explain what happens to the potential difference across R_2 as the resistance of R_1 is gradually increased from zero.

Mark Scheme (4 marks):

State what happens to the potential difference across R_2 :

- The potential difference across R_2 increases (as the potential difference across R_1 increases); *[1 mark]*

To explain, describe how the potential difference across R_1 changes:

- As the resistance of R_1 increases, the potential difference (V_1) across it increases; *[1 mark]*

Describe how the potential difference across R_1 affects the potential difference across R_3 :

- (Because of Kirchhoff's second law) the potential difference across R_3 = 5 V – potential difference across R_1

  AND

- Therefore, as the potential difference across R_1 increases, the potential difference across R_3 decreases; *[1 mark]*

Describe how the potential difference across R_3 affects the potential difference across R_2 :

- (Because of Kirchhoff's second law) the potential difference across R_2 = 6 V – potential difference across R_3

  AND

- Therefore as the potential difference across R_3 decreases, the potential difference across R_2 increases; *[1 mark]*

[Total: 4 marks]

This is a potential divider question in an unfamiliar context when you just consider the top loop. As R_1 increases in resistance, the potential difference across it increases. There is only a total e.m.f. of 5 V in this top loop, so less is available to be 'spent' on R_3.

Then, this becomes a potential divider question for the lower loop. If R_3 has a greater potential difference, there is less available for R_2 following the conservation of energy.

*The thought process for this question is simple, however this mark scheme is large because you needs to **explain** every step:*

- *If p.d. across R_1 increases then p.d. across R_3 decreases*
- *This is because the potential difference in the loop remains constant from V = IR*
- *So, p.d. across R_2 will also increase by the same logic*

Part (d) **[6 marks]**

Calculate the current flowing through each resistor when R_1 is set to a resistance of 10 Ω.

Mark Scheme (6 marks):

List the known quantities:

- R_1 = 10 Ω
- R_2 = 10 Ω
- R_3 = 30 Ω
- EMF_1, ε_1 = 5 V
- EMF_2, ε_2 = 6 V

Apply Kirchhoff's second law to the loop containing EMF_1:

- Kirchhoff's second law says the total EMF in a closed circuit (loop) equals the sum of potential differences across each component
- Therefore, $\epsilon_1 = V_1 + V_3$ (where V_1 and V_3 are the voltages across R_1 and R_3 respectively)
- Using Ohm's Law, $\epsilon_1 = I_1R_1 + I_3R_3$
- Therefore $5 = 10I_1 + 30I_3$ (equation 1) *[1 mark]*

Apply Kirchhoff's second law to the loop containing EMF_2:

- $\epsilon_2 = V_2 + V_3$ (where V_2 and V_3 are the voltages across R_2 and R_3 respectively)

- Again, using Ohm's Law, $\epsilon_2 = I_2R_2 + I_3R_3$
- Therefore $6 = 10I_2 + 30I_3$ (equation 2) *[1 mark]*

Substitute the expression from part (a) into equation 1 and equation 2:

- Since there are three unknowns, there needs to be three independent equations to solve simultaneously
- Therefore, from part (a), $I_1 + I_2 = I_3$ can be substituted for I_3 in equation 2 and equation 1
- Equation 2 becomes: $6 = 10I_2 + 30(I_1 + I_2) = 10I_2 + 30I_1 + 30I_2$
- Simplifying, $6 = 40I_2 + 30I_1$ (equation 3)

AND

- Equation 1 becomes: $5 = 10I_1 + 30(I_1 + I_2) = 10I_1 + 30I_1 + 30I_2$
- Simplifying, $5 = 30I_2 + 40I_1$ (equation 4) *[1 mark]*

Solve equation 3 and equation 4 simultaneously:

- (Equation 3) × 3 gives: $18 = 120I_2 + 90I_1$ (equation 5)
- (Equation 4) × 4 gives: $20 = 120I_2 + 160I_1$ (equation 6)
- (Equation 6) – (Equation 5) gives: $(20 - 18) = (120I_2 - 120I_2) + (160I_1 - 90I_1)$
- Therefore $2 = 70I_1$
- $I_1 = \frac{2}{70} = 0.029$ A (2 s.f.) *[1 mark]*

Substitute the value for I_1 into any equation to calculate I_2:

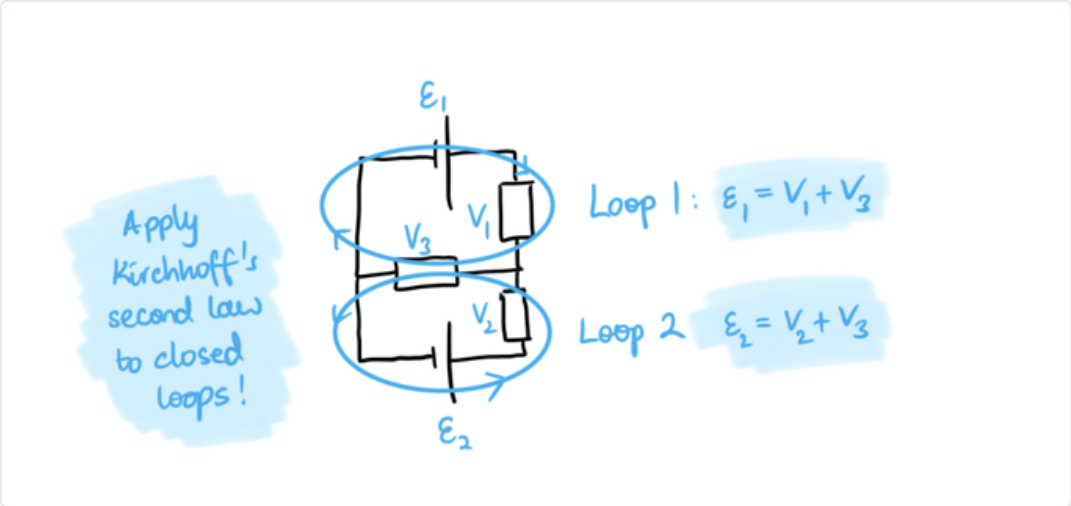
- Any equation can be used to calculate I_2 using the value for I_1
- Substituting $I_1 = 0.029$ into equation 4 gives: $5 = 30I_2 + 40(0.029)$
- $5 = 30I_2 + 1.16$
- $I_2 = \frac{5 - 1.16}{30} = 0.13$ A (2 s.f.) *[1 mark]*

Calculate I_3 using values for I_1 and I_2:

- Using the equation from part (a): $I_1 + I_2 = I_3$
- Therefore, $I_3 = 0.13 + 0.029 = 0.16$ A (2 s.f.) *[1 mark]*

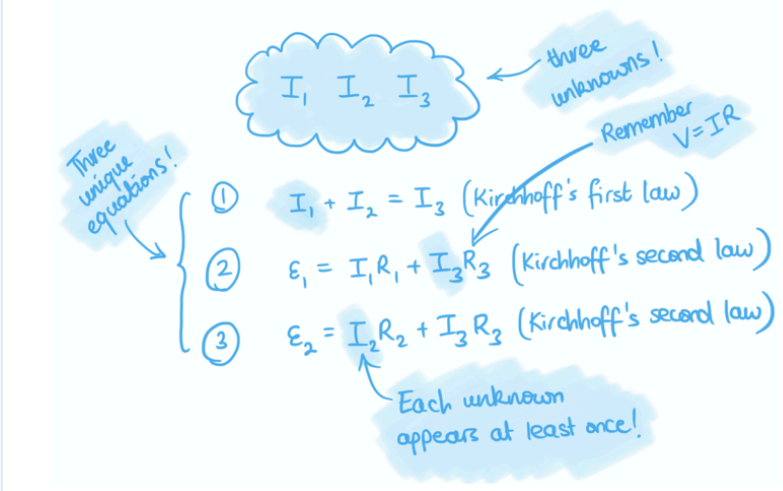
[Total: 6 marks]

This question is certainly very challenging; while circuits on the A Level paper are unlikely to require this level of analysis, there is nothing here outside the scope of your physics syllabus – notice, it only requires a very careful application of Kirchhoff's laws and Ohm's law. The image below shows how to apply Kirchhoff's second law to closed loops in a circuit:



Notice that the loops 'go in the direction' of conventional current in the loop, and also that EMFs (power supplies) are distinct to the potential differences across components (sometimes called voltage drops, because energy from the current is transferred away from it to the component itself).

Another key takeaway from this question, which is a very useful problem-solving tool to bear in mind generally, is that there are three unknowns – i.e., the currents I_1, I_2 and I_3. The number of unknown variables in any system tells you how many 'independent' (unique) equations you need to set up, in which every unknown appears at least once. So, in this question, we needed to set up ***three*** *independent equations; after we did this, it was just a case of some mathematical juggling to solve for each unknown. The image below recaps the three unknowns and the three independent equations we set up:*



There are of course multiple equivalent ways to solve simultaneous equations (depending on which variable you choose to eliminate first, and which equation to substitute into, etc.) but you should in principle be comfortable with solving simultaneous equations (two for two unknowns and three for three unknowns!). Have a look at the Save My Exam GCSE and / or A-level maths notes to refresh your knowledge on these if you are stuck!

Question 9 Hard [9 marks]

Part (a) [2 marks]

Fig. 1.1 shows a potentiometer in action. The wire in the potentiometer is represented by a fixed resistor of resistance 10 Ω and the sliding contact C can move from one end of the wire (at position P) to the other end of the wire (position Q). This allows the output potential difference (p.d.) to be varied. The potentiometer is in series with another fixed resistor of 5 Ω.

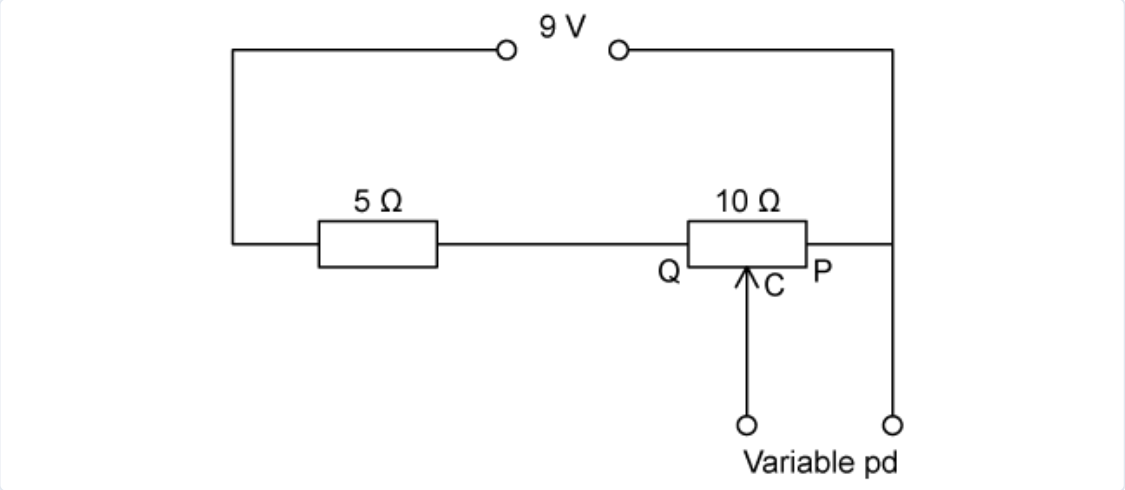


Fig. 1.1

This circuit is powered by a 9 V power supply and is used to supply a variable potential difference to another circuit.

Sketch a graph on the axes provided in Fig. 1.2 to show how the supplied potential difference *V* varies as the moving contact C is moved from position P to position Q.

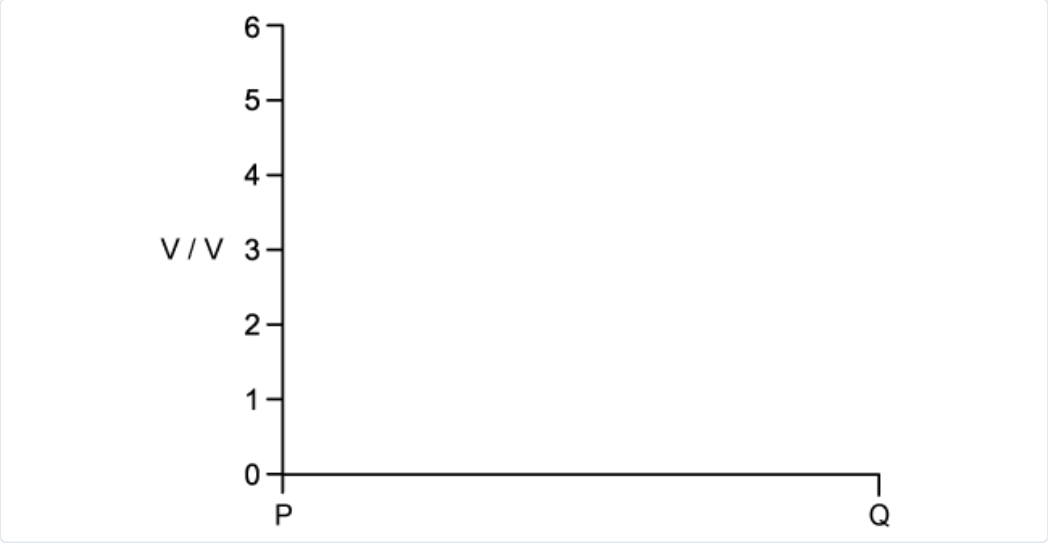
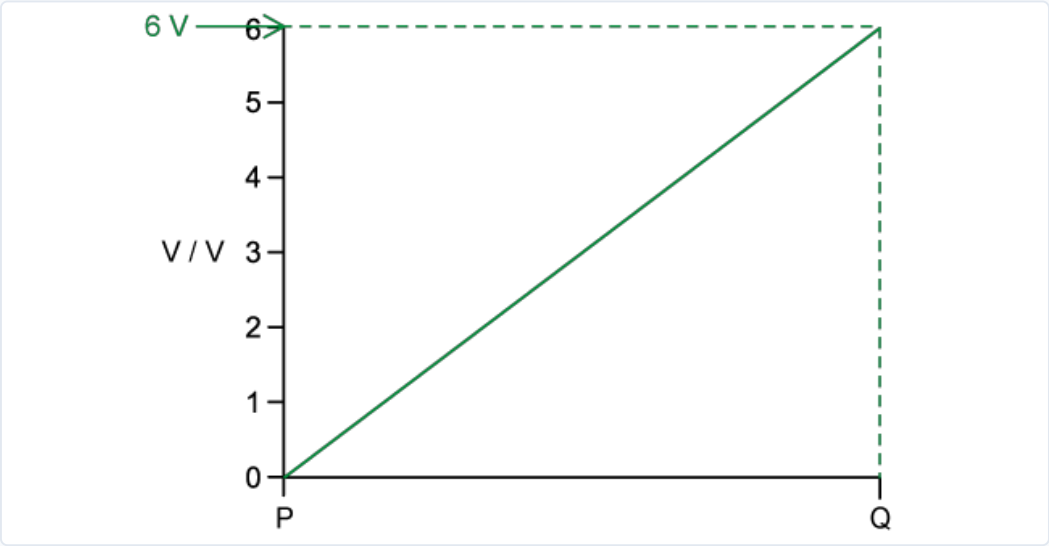


Fig. 1.2

Sketch a graph on the axes provided in Fig. 1.2 to show how the supplied potential difference *V* varies as the moving contact C is moved from position P to position Q.

Mark Scheme (2 marks):

- A graph which would score 2 marks should look like:

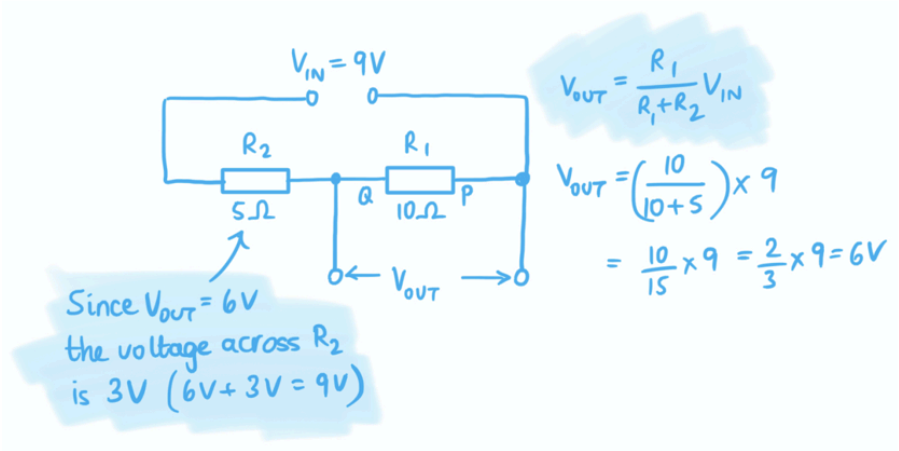


- Straight line with a positive gradient; *[1 mark]*
- At P, V = 0 V **AND** at Q, V = 6 V *[1 mark]*

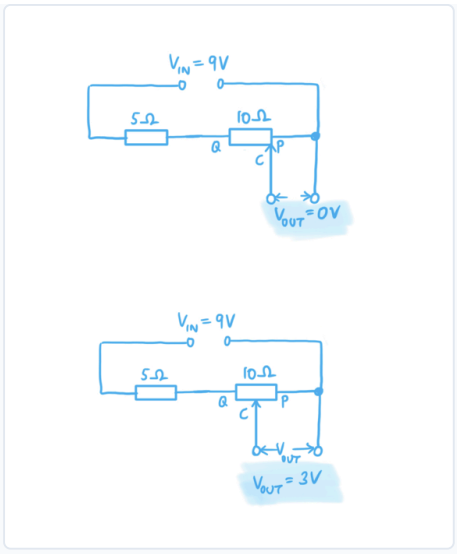
[Total: 2 marks]

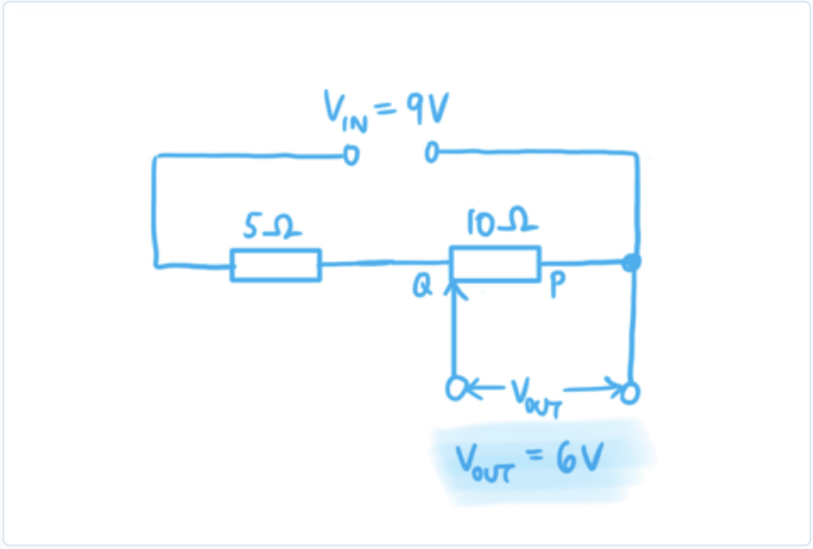
This circuit is a potential divider, because there are two resistors in series sharing the same EMF. As the slider moves, the potential is 'divided' between them.

By inspection, the resistances are in the ratio 1:2, so the potential is also split in the ratio 1:2 (with the bigger share of voltage across the larger resistance). Hence, the 9 V EMF is divided as 3 V and 6 V. This can be seen more explicitly using the potential divider equation, shown below:



Note that if the sliding contact C in Fig. 1.1 is at position P, then there is no potential difference across the variable pd (the voltage supplied is 0 V). The output pd (in parallel with the resistor) increases as more of the resistor is included between one end (position P) and the sliding C. This is illustrated below:





Therefore, the potentiometer can provide a variable potential difference output between 0 V and 6 V.

Part (b) **[4 marks]**

Potential divider circuits generally comprise of two or more resistors in series with each other.

Some potential divider circuits have multiple branches connected in parallel, such as that shown in Fig. 1.3:

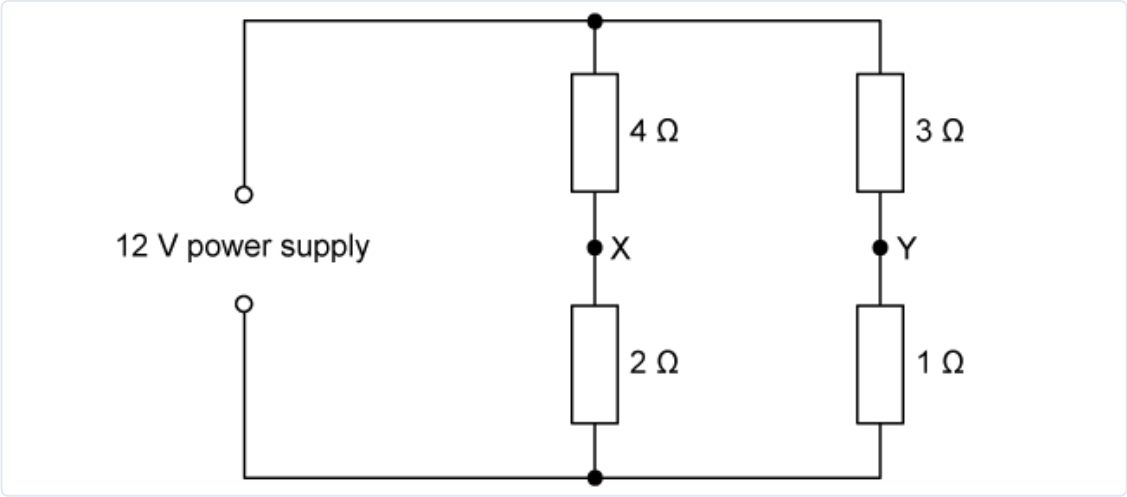


Fig. 1.3

Show that the potential difference between X and Y, V_{OUT}, is 1 V.

Mark Scheme (4 marks):

List the known quantities:

- V_{IN} = 12 V
- Resistor R_1 = 4 Ω
- Resistor R_2 = 2 Ω
- Resistor R_3 = 3 Ω
- Resistor R_4 = 1 Ω

Method 1 (clockwise current)

Determine the potential difference across R_1 :

- The potential divider equation for resistor R_1 (which is in a series loop with resistor R_2) is $V_{OUT} = \frac{R_1}{R_1 + R_2} V_{IN}$
- Therefore p.d. across R_1 is $V_1 = \frac{4}{4+2} \times 12 = \frac{4}{6} \times 12 = 8$ V *[1 mark]*

Determine the potential difference across R_3 :

- The potential divider equation for resistor R_3 (which is in a series loop with resistor R_4) is $V_{OUT} = \frac{R_3}{R_3 + R_4} V_{IN}$
- Therefore p.d. across R_3 is $V_3 = \frac{3}{3+1} \times 12 = \frac{3}{4} \times 12 = 9$ V *[1 mark]*

Determine the potential at X and Y:

- Charge transfers 8 V across resistors R_1
- The potential at X is therefore 12 V – 8 V = 4 V

AND

- Similarly, charge transfers 9 V of potential across resistor R_3
- The potential at Y is therefore 12 V – 9 V = 3 V *[1 mark]*

Show the potential difference between X and Y is 1 V:

- The potential difference between X and Y is therefore V_{OUT} = 4 V – 3 V = 1 V *[1 mark]*

Method 2 (anticlockwise current)

Determine the potential difference across R_2 :

- The potential divider equation for resistor R_2 (which is in a series loop with resistor R_1) is $V_{OUT} = \frac{R_2}{R_2 + R_1} V_{IN}$
- Therefore p.d. across R_2 is $V_2 = \frac{2}{2+4} \times 12 = \frac{2}{6} \times 12 = 4$ V *[1 mark]*

Determine the potential difference across R_4 :

- The potential divider equation for resistor R_4 (which is in a series loop with resistor R_3) is $V_{OUT} = \frac{R_4}{R_4 + R_3} V_{IN}$
- Therefore p.d. across R_4 is $V_4 = \frac{1}{1+3} \times 12 = \frac{1}{4} \times 12 = 3$ V *[1 mark]*

Determine the potential at X and Y:

- Charge transfers 4 V across resistor R_2
- The potential at X is therefore 12 V – 4 V = 8 V

AND

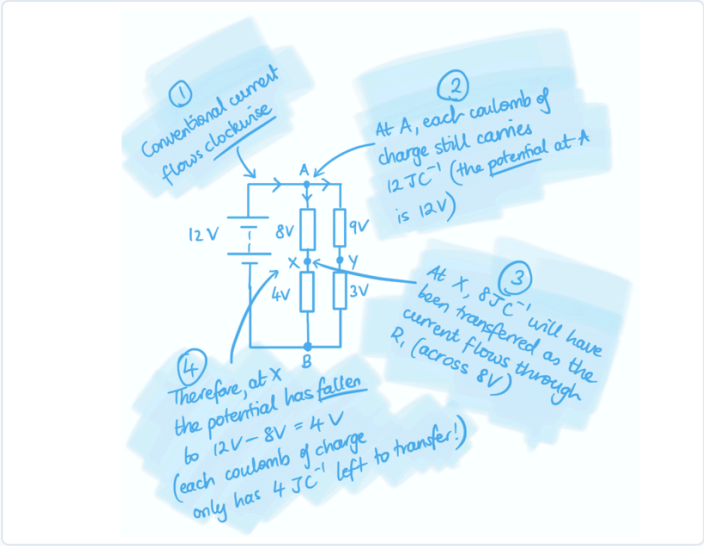
- Charge transfers 3 V across resistor R_4
- The potential at X is therefore 12 V – 3 V = 9 V *[1 mark]*

Show the potential difference between X and Y is 1 V:

- The potential difference between X and Y is therefore V_{OUT} = 9 V – 8 V = 1 V *[1 mark]*

[Total: 4 marks]

Method 1 *assumes that the current flows in a clockwise direction through the circuit: in other words, that the terminals of the power supply are oriented as shown in the sketch below. You should work your way through the four steps, to convince yourself that the potential at point X is 4V* ***if the current flows clockwise*** *through the circuit.*



Following this reasoning, you should be able to check and see that the potential at Y is 12 V – 9 V = 3 V, so the potential ***difference*** *between X and Y is 1 V.*

Method 2 *assumes the current flows in an anti-clockwise direction through the circuit; in that case, simply reverse the direction of the current in the illustration above and you should see that the potential at X is 12 V – 4 V = 8 V. Similarly, the potential at Y is 12 V – 3 V = 9 V, giving a potential difference between X and Y as 1 V again.*

Clearly, either assumption is valid and will give the correct ***potential difference*** *between* ***X*** *and* ***Y****. This question introduces you to a fact you will explore more in A Level physics: 'potential' as a quantity is relative (i.e., it depends on how you define directions), but potential difference is always absolute – so long as your working is consistent!*

Part (c) [3 marks]

Fig. 1.4 shows another potential divider circuit which includes a thermistor with resistance *R*.

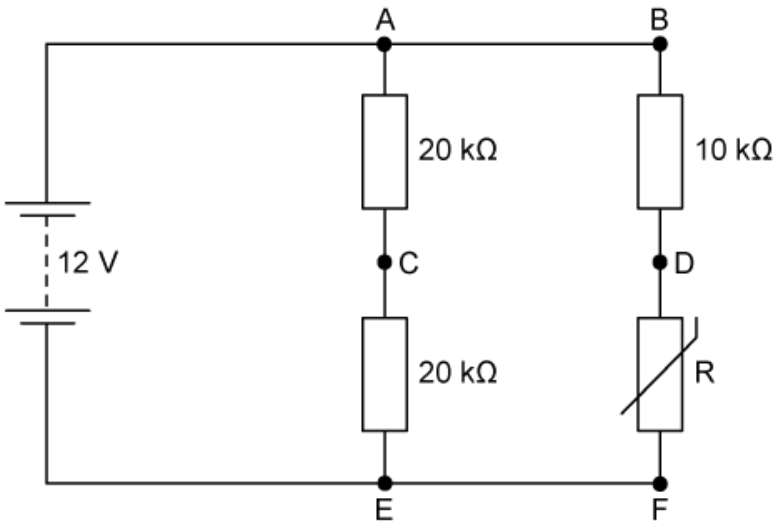


Fig. 1.4

The battery has an EMF of 12 V, with negligible internal resistance. At room temperature, the resistance of the thermistor is 5 kΩ.

Calculate the current in the battery at room temperature.

Mark Scheme (3 marks):

List the known quantities:

- Resistance of the thermistor (at room temperature), R = 5 kΩ = 5000 Ω
- Resistance between AC = 20 kΩ = 20 000 Ω
- Resistance between CE = 20 kΩ = 20 000 Ω
- Resistance between BD = 10 kΩ = 10 000 Ω
- EMF of the battery = 12 V

Determine the series resistances:

- Resistors in series $R_T = R_1 + R_2 + R_3 + \cdots$
- There are two resistors in series between A and E, so $R_T = R_1 + R_2$
- Therefore, the total resistance between A and E = resistance between AC + resistance between CE = 20 000 + 20 000 = 40 000 Ω

  AND

- There are two resistors in series between B and F, so $R_T = R_1 + R_2$
- Therefore, the total resistance between B and F = resistance between BD + resistance of the thermistor (at room temperature) = 10 000 + 5000 = 15 000 Ω *[1 mark]*

Determine the total resistance of the circuit:

- Resistors in parallel $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots$
- The resistance between A and E is in parallel to the resistance between B and F
- Therefore $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{40\ 000} + \frac{1}{15\ 000} = \frac{11}{120\ 000}$
- Therefore $R_T = \frac{120\ 000}{11}$ = 10 909 Ω *[1 mark]*

Determine the current through the battery:

- Using Ohm's law across the battery, $V = IR$
- V is the EMF ϵ of the cell
- Therefore 12 = I × 10 909
- $I = \frac{12}{10\ 909}$ = 1.1 × 10^{-3} A *[1 mark]*

[Total: 3 marks]